

[DM] PROJECT - QUERIES

Query 1 - Recompute the Basic Cube.

```
SELECT      x1.manager_name as "Home Coach", t1.team_name as "Home Team",
            m.home_goals as "Home Goals",
            x2.manager_name as "Away Coach", t2.team_name as "Away Team",
            m.away_goals as "Away Goals",
            d.day as "Date", c.competition_name as "League", c.type as "Type",
            r.referee_name as "Referee", y.season_year as "Season", m.attendance as "Visitor Count",
            c.country as "Country", c.country_population as "Population",
            s.stadium_name as "Stadium", s.city as "City", s.capacity as "Capacity",
            p1.position as "Home Rank", p2.position as "Away Rank", t3.team_name as "League Winner",
            t1.ucl_won as "UCL Home", t1.uel_won as "UEL Home",
            t1.cwc_won as "CWC Home", t1.usc_won as "USC Home",
            t2.ucl_won as "UCL Away", t2.uel_won as "UEL Away",
            t2.cwc_won as "CWC Away", t2.usc_won as "USC Away"

FROM        match m
            JOIN team t1 ON m.key_home = t1.key_team
            JOIN team t2 ON m.key_away = t2.key_team
            JOIN date d ON m.key_date = d.key_date
            JOIN competition c ON m.key_competition = c.key_competition
            LEFT JOIN referee r ON m.key_referee = r.key_referee
            LEFT JOIN stadium s ON m.key_stadium = s.key_stadium
            JOIN season y ON d.key_season = y.key_season
            JOIN manager x1 ON (d.key_date = x1.key_date AND t1.key_team = x1.key_team)
            JOIN manager x2 ON (d.key_date = x2.key_date AND t2.key_team = x2.key_team)
            JOIN position p1 ON (y.key_season = p1.key_season AND t1.key_team = p1.key_team)
            JOIN position p2 ON (y.key_season = p2.key_season AND t2.key_team = p2.key_team)
            JOIN winner w ON (c.key_competition = w.key_competition AND y.key_season = w.key_season)
            JOIN team t3 ON t3.key_team = w.key_winner_team
```

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Query 2 - OLAP Operators.

ROLL-UP:

```
SELECT      y.season_year,
            t.team_name,
            c.competition_name,
            SUM(m.attendance) as "Total Attendance"

FROM        match m
            JOIN team t ON m.key_home = t.key_team
            JOIN date d ON m.key_date = d.key_date
            JOIN competition c ON m.key_competition = c.key_competition
            JOIN season y ON d.key_season = y.key_season

GROUP BY    y.season_year, t.team_name, c.competition_name

ORDER BY    y.season_year, c.competition_Name, "Total Attendance" DESC
```

DRILL-DOWN:

```
SELECT      d.month,
            t.team_name,
            c.competition_name,
            SUM(m.attendance) as "Monthly Attendance"

FROM        match m
            JOIN team t ON m.key_home = t.key_team
            JOIN date d ON m.key_date = d.key_date
            JOIN competition c ON m.key_competition = c.key_competition

GROUP BY    d.month, t.team_name, c.competition_name

ORDER BY    d.month, c.competition_Name, "Monthly Attendance" DESC
```

SLICE-AND-DICE:

```
SELECT      y.season_year as "Season",
            t1.team_name as "Home Team",
            t2.team_name as "Away Team",
            c.competition_name as "League",
            m.home_goals as "Home Goals",
            m.away_goals as "Away Goals",
            m.attendance as "Attendance"

FROM        match m
            JOIN team t1 ON t1.key_team = m.key_home
            JOIN team t2 ON t2.key_team = m.key_away
            JOIN date d ON d.key_date = m.key_date
            JOIN season y ON y.key_season = d.key_season
            JOIN competition c ON c.key_competition = m.key_competition

WHERE       c.competition_name = 'Serie A' AND
            m.attendance > 30.000 AND
            (m.home_goals + m.away_goals) >= 2

ORDER BY    season_year, "Attendance" DESC
```

Query 3 - Select the name of the teams, the competition, the score and the date of all those matches played in 2018 by West Ham, ended up with a win or a draw.

```
SELECT      t1.team_name as "Home", t2.team_name as "Away",
            c.competition_name, m.home_goals, m.away_goals,
            d.day

FROM        match m JOIN team t1 ON m.key_home = t1.key_team
            JOIN team t2 ON m.key_away = t2.key_team
            JOIN competition c ON m.key_competition = c.key_competition
            JOIN date d ON m.key_date = d.key_date

WHERE       ((t1.team_name = 'West Ham' AND m.home_goals >= m.away_goals) OR
            (t2.team_name = 'West Ham' AND m.away_goals >= m.home_goals))
            AND d.year = 2018
```

Query 4 - Which stadiums had the highest annual average attendance in Ligue 1 matches after 2015?

```
SELECT      s.stadium_name AS "Stadium",
            s.city AS "City",
            ROUND(AVG(m.attendance)) AS "Avg Attendance",
            y.season_year AS "Year",
            COUNT(*) AS "# Matches"

FROM        match m
            JOIN competition c ON m.key_competition = c.key_competition
            JOIN date d ON m.key_date = d.key_date
            JOIN stadium s ON m.key_stadium = s.key_stadium
            JOIN season y ON d.key_season = y.key_season

WHERE       c.competition_name = 'Ligue 1' AND
            d.year >= 2015

GROUP BY    s.stadium_name, s.city, y.season_year

HAVING      COUNT(*) > 10

ORDER BY    "Avg Attendance" DESC
```

Query 5 - Find referees who officiated more than 10 Serie A matches, and compute average goals per match.

```
SELECT      r.referee_name AS "Referee",
            COUNT(*) AS "# Matches",
            ROUND(AVG(m.home_goals + m.away_goals), 2) AS "Avg Goals"

FROM        match m
            JOIN referee r ON m.key_referee = r.key_referee
            JOIN competition c ON m.key_competition = c.key_competition

WHERE       c.competition_name = 'Serie A'

GROUP BY    r.referee_name

HAVING      COUNT(*) > 10

ORDER BY    "Avg Goals" DESC
```